Quiz 1 ver. 1

Discrete Structures

4th day of September of the year of our Lord 2026

Answer each of the following questions according to their instructions.

For all of the following problems, fix a universe of discourse containing all of the natural numbers 0, 1, 2, … and so on. Let $<$, $>$, and $=$ be the binary predicates for *"less than," "greater than,"* and *"equality"* respectively. Recall that we use the symbols $+$ and $\cdot$ to denote *"addition"* and *"multiplication"* of natural numbers.

1. We say that a natural number x is *"even"* when it is a multiple of 2, which means we can find some natural number y with the property $2 \cdot y = x$. Similarly, we call a natural number *"odd"* when it's one more than a multiple of 2.

   a. Write a well-formed formula with one free variable x that says *"x is even."*

   $$\exists y(2 \cdot y = x)$$

   b. Write a proposition that says *"None of the even numbers are odd."*

   $$\forall x\big(\exists y(2 \cdot y = x) \to \neg\exists y(2 \cdot y = x + 1)\big)$$

2. Answer each of the following questions either *True* or *False* but not both.

   a. $\forall x \forall y(x < y)$.

   ○ *True*
   ● *False*

   b. $\forall x\big(x = 0 \to \forall y(x \cdot y = 0)\big)$.

   ● *True*
   ○ *False*